

Customer Churn Prediction

A machine learning project that predicts which telecom customers are likely to leave, so the business can retain them before they go. Built end to end with Random Forest and XGBoost, class balancing with SMOTE, and a self-driven feature experiment.

BEST ROC-AUC	CHURNERS CAUGHT	DATASET	MODELS
0.821	~60%	7,043	2
Random Forest	on unseen test set	customers, 21 fields	RF & XGBoost

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Datheon Machine Learning
& Data Science Internship
Task 3

SECTION 1

Business Problem & Objective

A telecom company loses revenue every time a customer leaves. Acquiring a new customer costs far more than keeping an existing one, so the company wants to identify at-risk customers **in advance**, and understand **why** they leave, so retention effort can be focused where it matters most.

- **Business question:** Which customers are likely to churn, and what drives it?
- **Data science task:** Supervised binary classification (Churn = Yes / No).
- **Success metric:** ROC-AUC above 0.80 on an unseen test set.

The dataset contains **7,043 customers** described by demographics, subscribed services, contract and billing details, and whether they churned. The target is imbalanced: about **27% churned and 73% stayed**, which shapes every modeling decision that follows.

SECTION 2

Exploratory Data Analysis

The exploratory analysis suggests that the company may face challenges in retaining new customers, particularly those in the higher monthly-charge group. Churn rates are considerably higher among customers with shorter tenure and decrease as tenure increases. This may indicate that the initial customer experience, pricing, or perceived value could influence retention. In contrast, customers with longer tenure may have established relationships with the company and may be more accustomed to its services. However, further analysis of customer satisfaction, contract type, and other factors is needed before concluding that the company has an ineffective retention strategy.

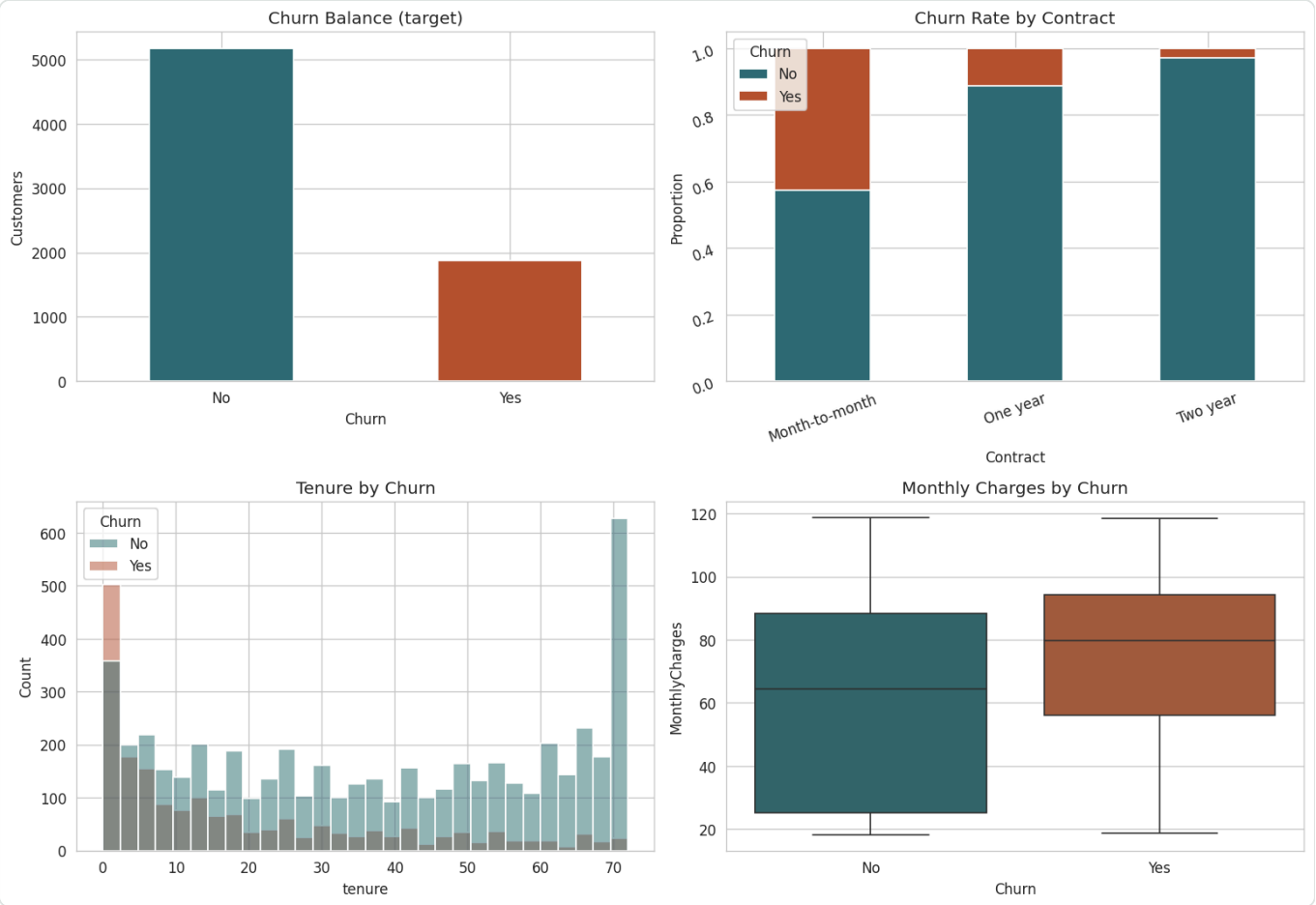


Figure 1. Churn balance, churn rate by contract, tenure distribution, and monthly charges by churn.

- **Contract type is the strongest single signal.** Month-to-month customers churn around 42%; one-year falls to roughly 11%; two-year is almost negligible.
- **New customers churn most.** Churn is concentrated in the first year and falls sharply with tenure.
- **Higher monthly charges** are associated with higher churn.

Interaction effect: tenure and monthly charges

To test whether two factors combine, customers were grouped by tenure band and monthly-charge band, and the churn rate measured in each cell. The result confirms an **interaction**: new customers who also pay high charges are the single most at-risk group, more than either factor alone would suggest.

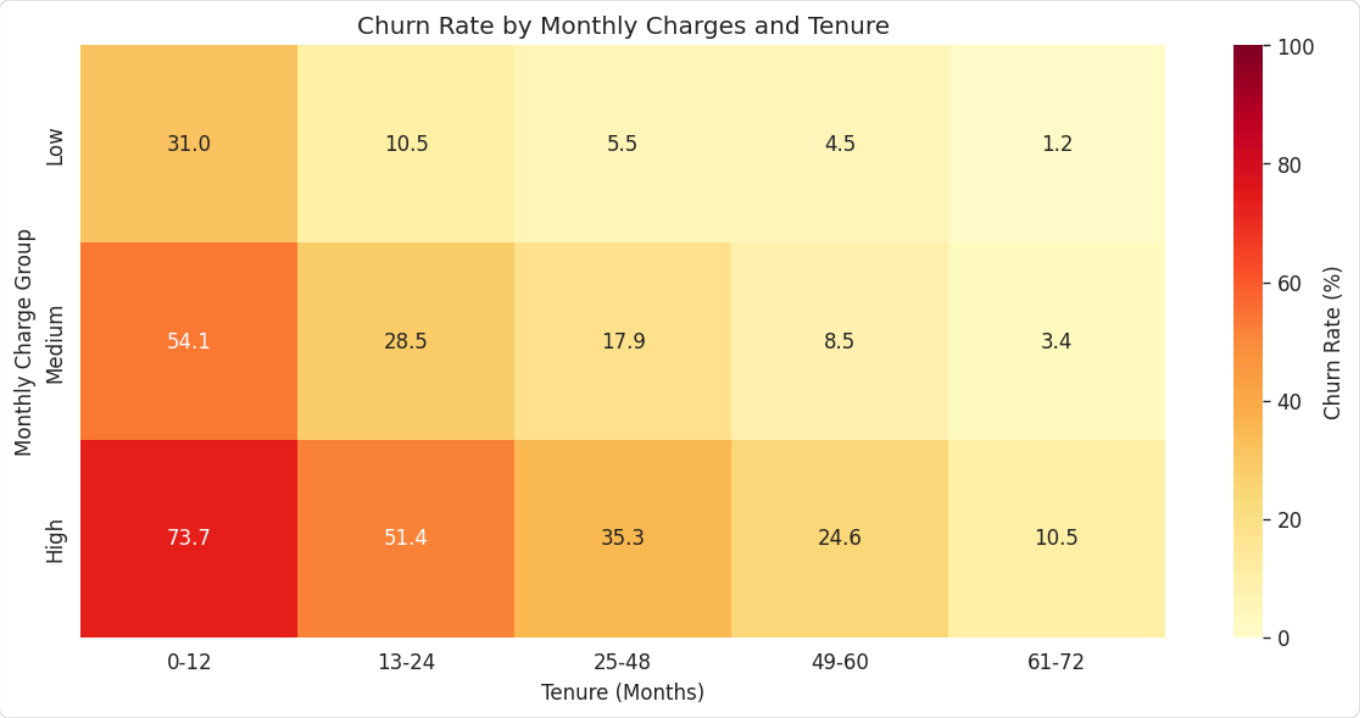


Figure 2. Churn rate (%) by monthly-charge group and tenure band.

SECTION 3

Methodology

Data preparation

- **Fixed TotalCharges:** stored as text with 11 blanks for brand-new customers (tenure 0); converted to numeric and filled with 0.
- **Dropped customerID:** an identifier with no predictive value.
- **Encoded the target:** Churn Yes = 1, No = 0.
- **One-hot encoded** categorical features, expanding 20 columns to 30 numeric features without imposing a false order.

Train / test split

An 80/20 split, stratified on Churn to preserve the ~27% churn ratio in both sets, with a fixed random seed for reproducibility. The test set was held out **before any balancing** to prevent data leakage.

Handling class imbalance (SMOTE)

SMOTE was applied to the **training set only**, synthesising minority-class examples until the classes were balanced. The test set was never altered, so evaluation reflects the real world.

Stage	Stayed	Churned
Before SMOTE	4,139	1,495
After SMOTE	4,139	4,139

SECTION 4

Results

Two ensemble models were trained on the balanced data and evaluated on the untouched test set.

Model	ROC-AUC	Churn Recall	Churn Precision
Random Forest (final)	0.821	0.60	0.58
XGBoost	0.819	0.60	0.56

Both models cleared the 0.80 target. Random Forest performed marginally better and was chosen as the final model; the difference is within noise and is reported honestly rather than overstated. The final model catches about **60% of real churners**. Because a retention offer costs far less than a lost customer, recall was prioritised over precision.

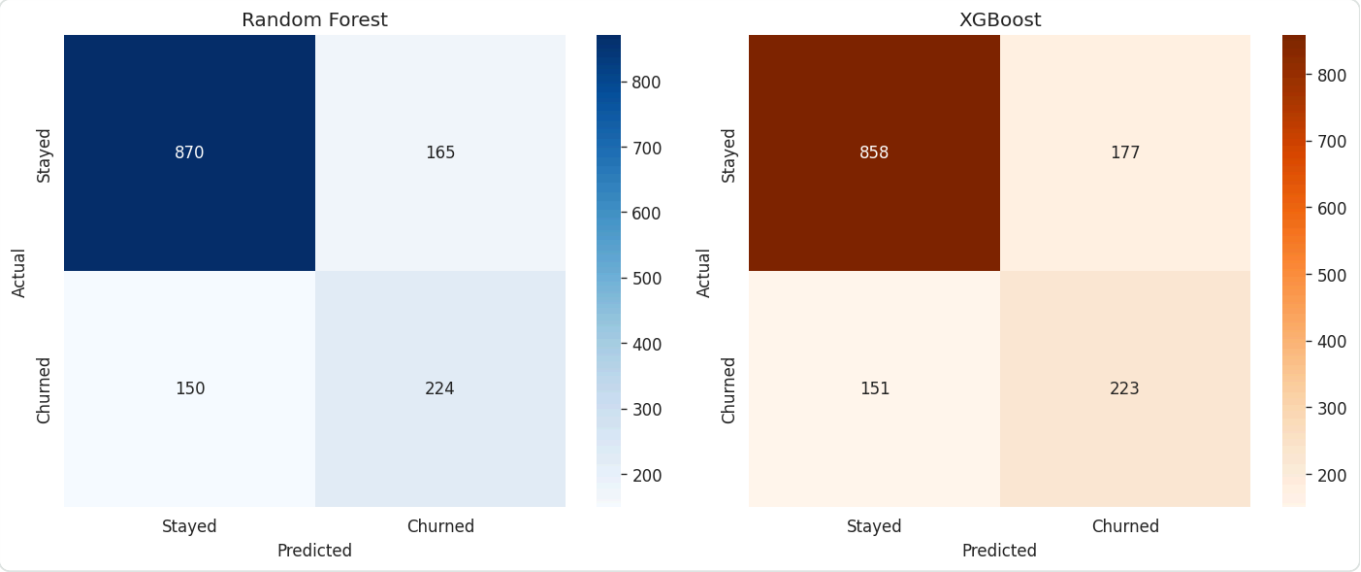


Figure 3. Confusion matrices for Random Forest and XGBoost on the test set.

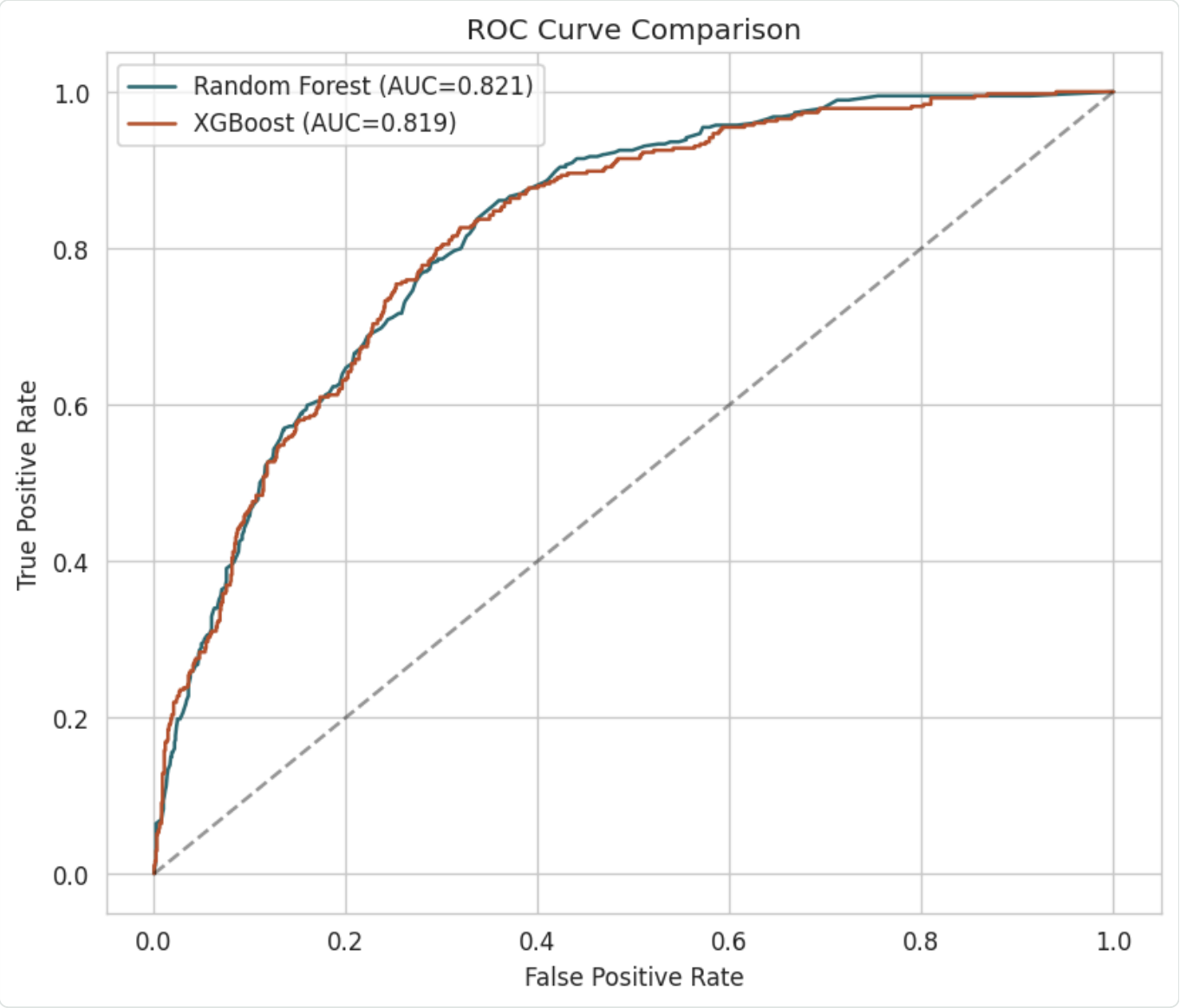


Figure 4. ROC curve comparison. Both models separate churners from stayers about 82% of the time.

SECTION 5

Interpretation

The most influential features align with the exploratory findings: **tenure, total and monthly charges, electronic-check payment, fiber-optic internet, and contract type**. The model's view of what matters matches the human-readable patterns, which builds confidence in the result.

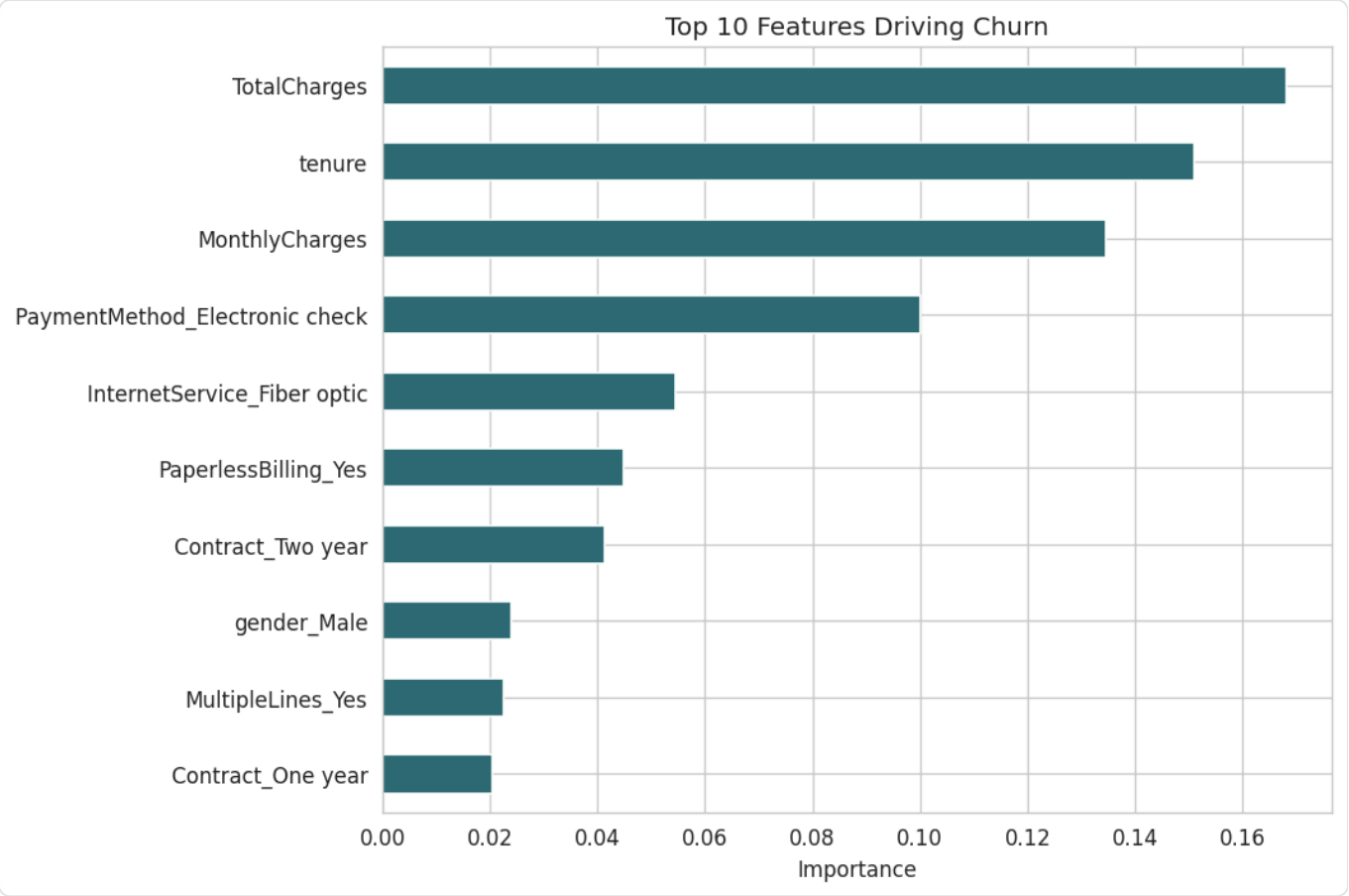


Figure 5. Top 10 features driving churn (Random Forest importance).

BEYOND THE BRIEF

Feature Experiment

HYPOTHESIS TESTED

The task only asked for a prediction model. As a data scientist, I wanted to go further: do we really need all 30 features, or can a leaner model do nearly as well?

The model was trained on the top 5, 10, 15, 20, 25, and 30 ranked features, and ROC-AUC compared at each step.

Features used	5	10	15	20	30
ROC-AUC	0.782	0.799	0.815	0.818	0.821

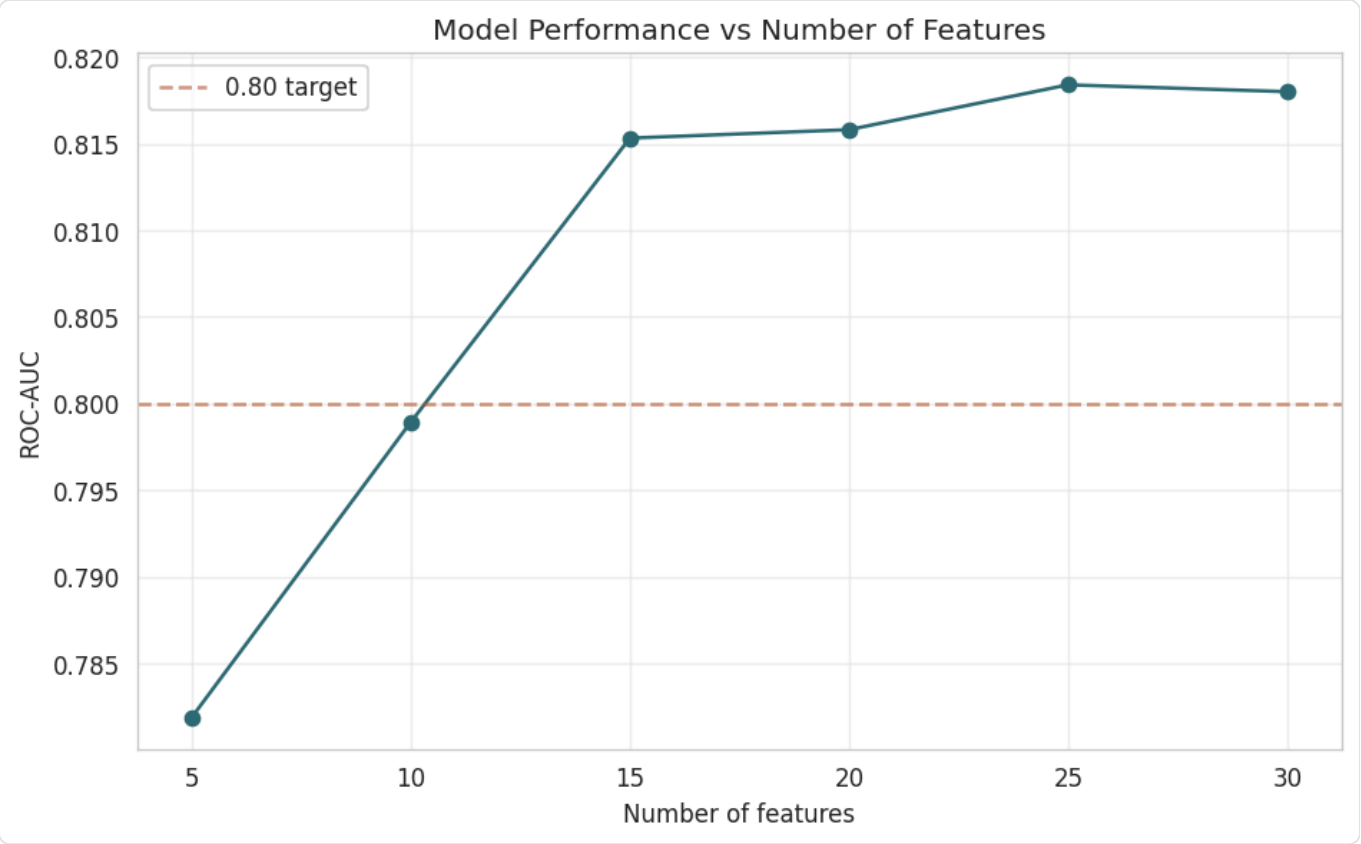


Figure 6. Model performance versus number of features. Performance plateaus after about 15 features.

Finding: performance rises steeply up to about 15 features, then flattens. The model reaches roughly 98% of its full performance with half the features, a clear illustration of the trade-off between simplicity and accuracy. The full model was kept as final for maximum performance, with this experiment documented as supporting analysis.

SECTION 6

Business Recommendations

1. **Target new customers in their first year** with onboarding and early check-ins. This is the highest-risk window.
2. **Move month-to-month customers onto annual contracts** with incentives. Contract length is the strongest protective factor.
3. **Investigate the electronic-check and fiber-optic segments** for pricing or service-quality issues.
4. **Focus retention budget on high-charge, low-tenure customers**, the group the interaction analysis flagged as most at risk.

SECTION 7

Conclusion & Tools

This project delivered a churn model that exceeds the target (ROC-AUC 0.82) and, more importantly, translates predictions into concrete retention actions. Beyond the required deliverable, a feature experiment demonstrated how much of the model's power comes from a small set of features, showing analytical depth rather than stopping at a single score.

Tools

Python, pandas, NumPy, scikit-learn, imbalanced-learn (SMOTE), XGBoost, Matplotlib, Seaborn.
